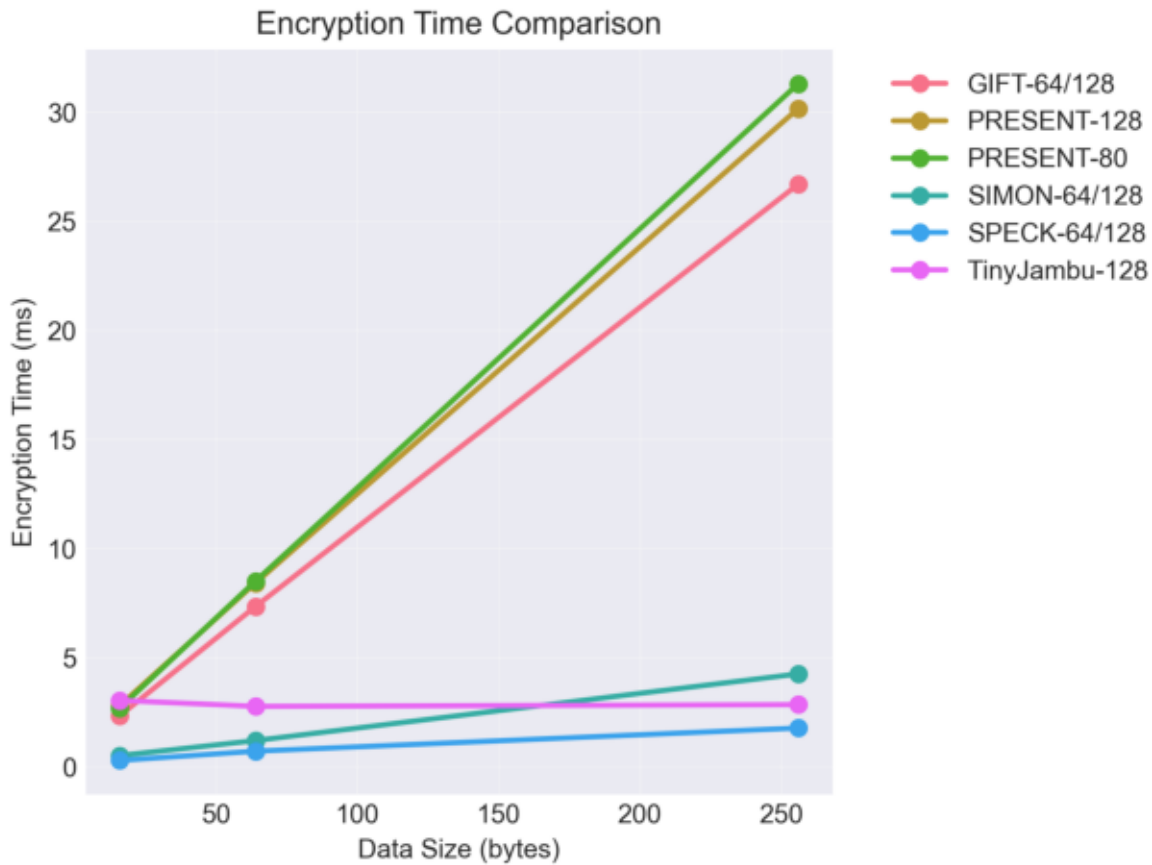
IoT Cryptography Project

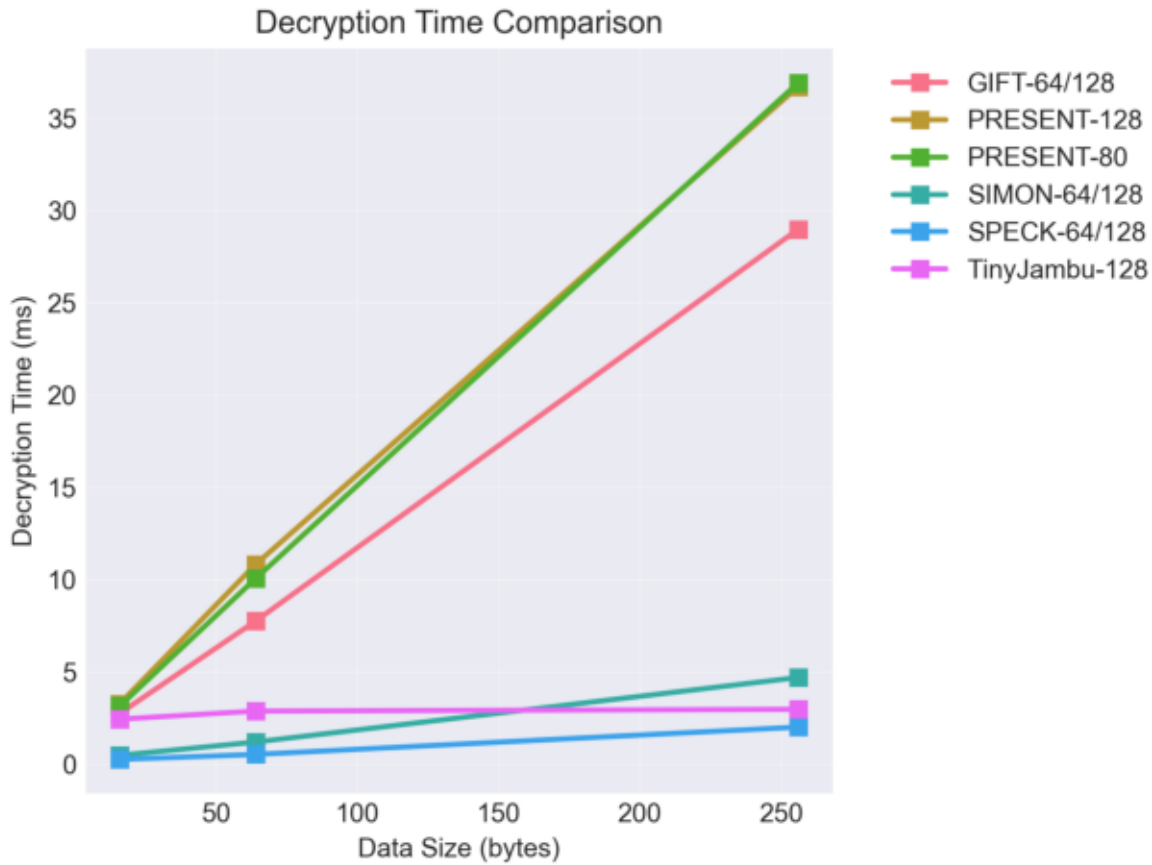
Performance & Security Analysis

Generated: 2026-04-06 13:24:51

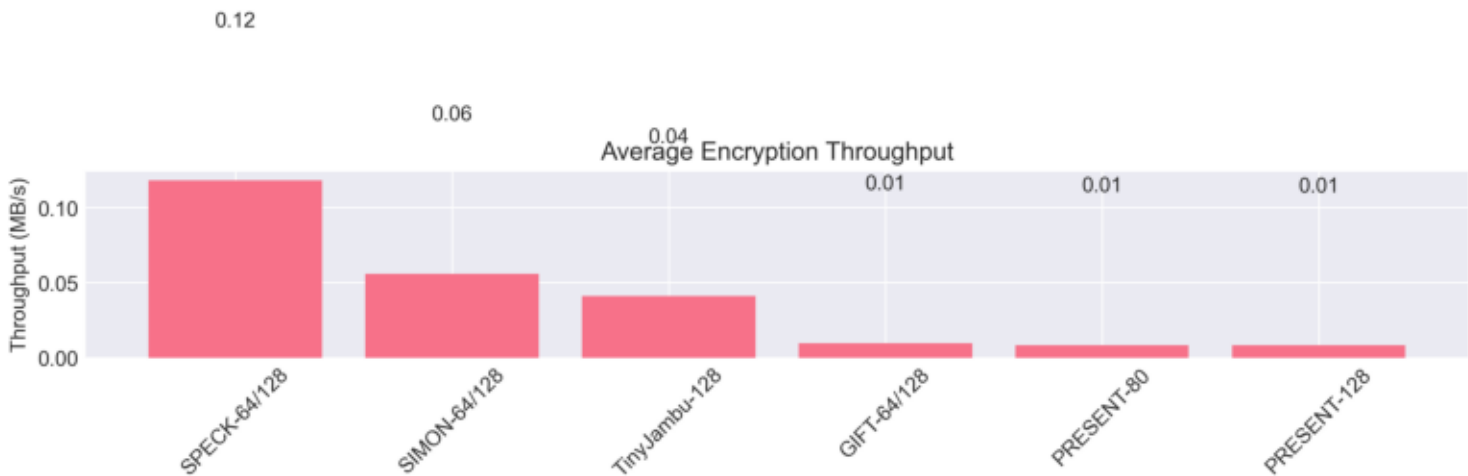
Encryption Time



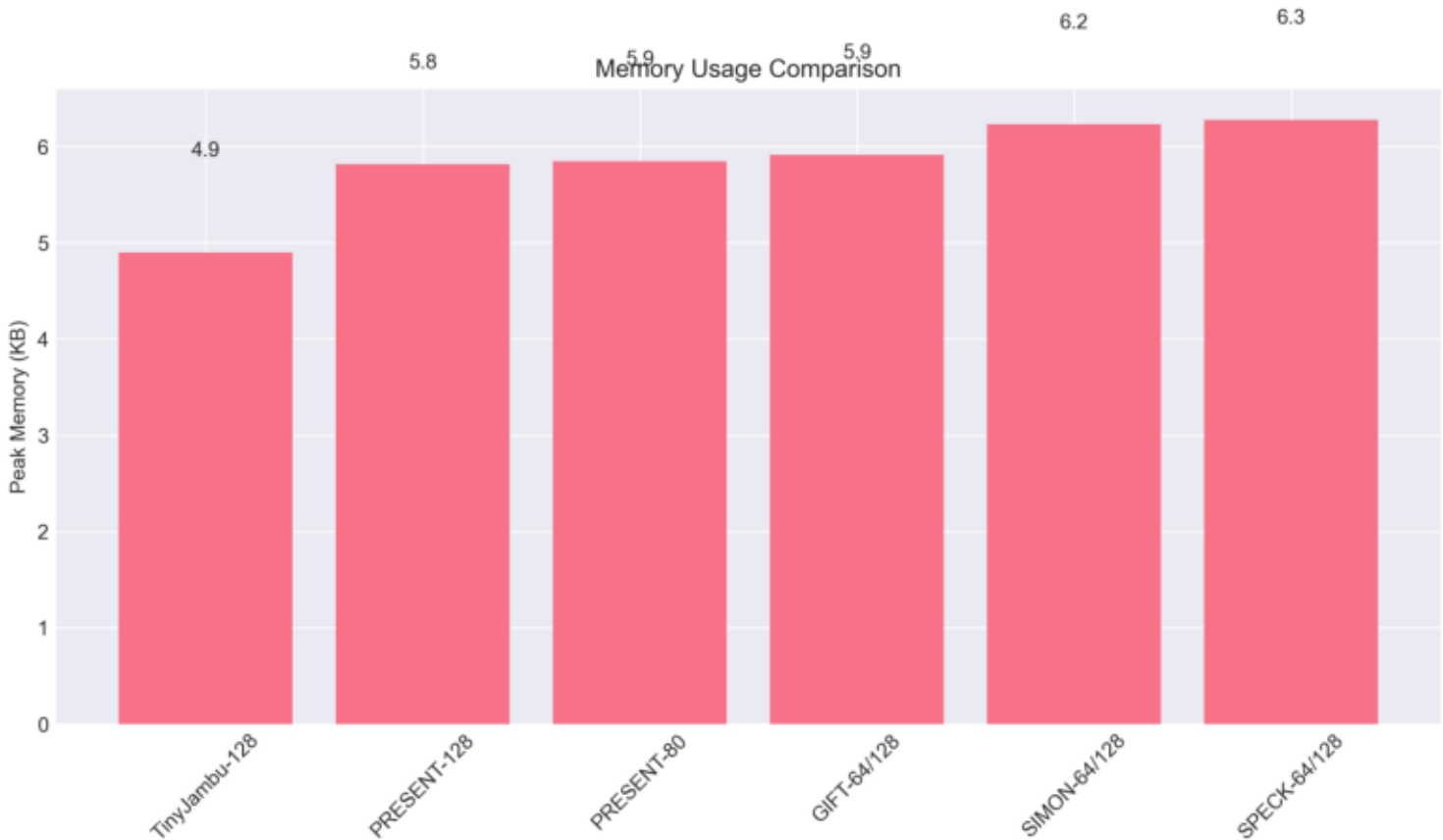
Decryption Time



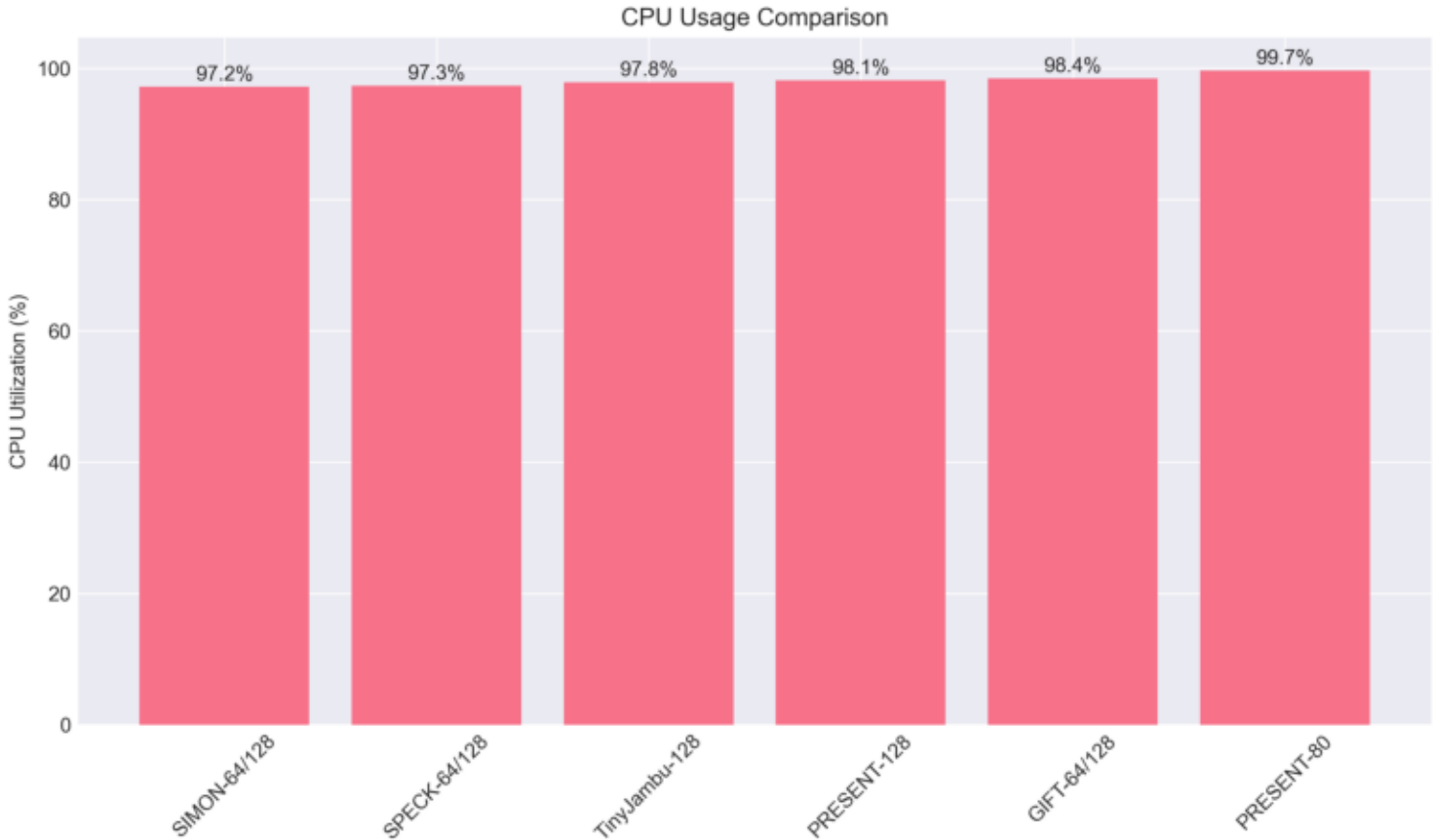
Throughput


Average Encryption Throughput
Throughput (MB/s)
0.10
0.05
0.00
0.12
0.06
0.04
0.01
0.01
0.01
SPECK-64/128
SIMON-64/128
TinyJambu-128
GIFT-64/128
PRESENT-80
PRESENT-128

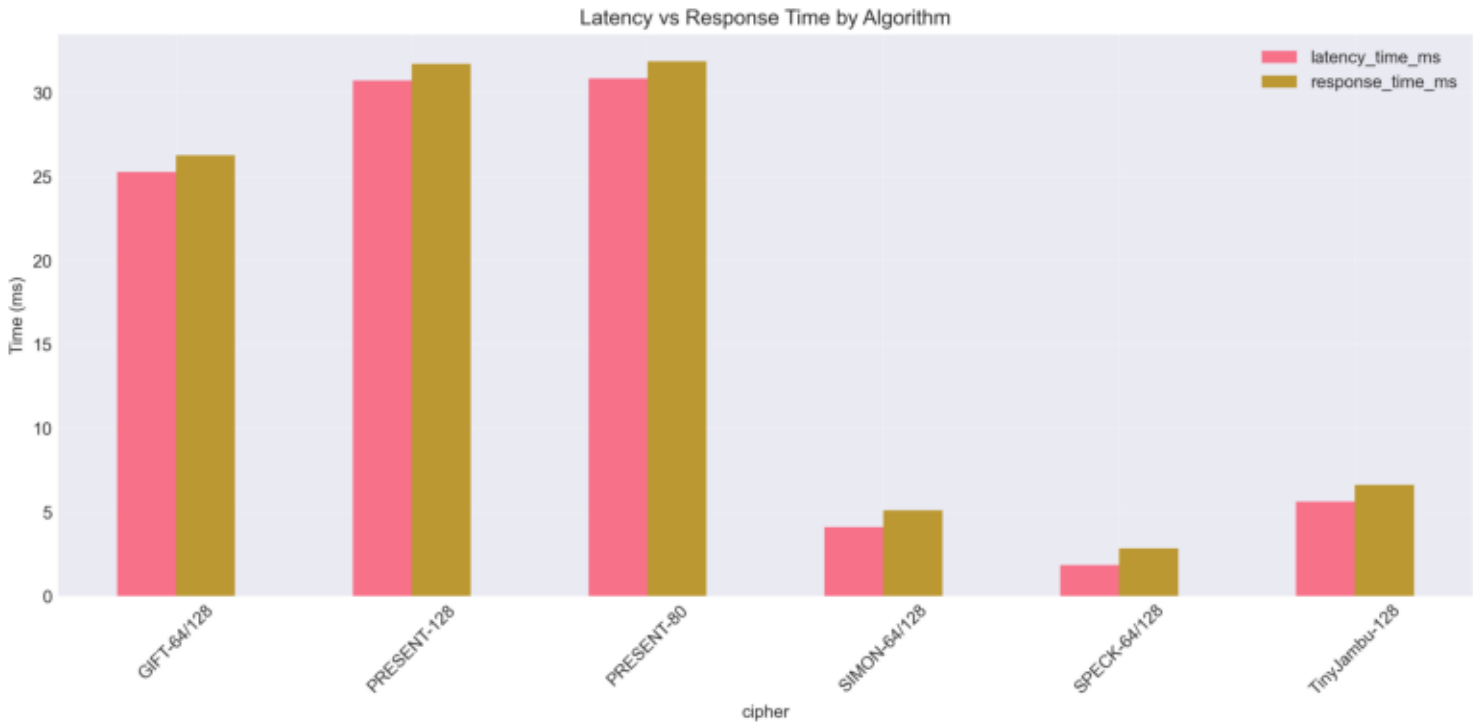
Memory


Memory Usage Comparison
Peak Memory (KB)
0
1
2
3
4
5
6
4.9
5.8
5.9
5.9
6.2
6.3
TinyJambu-128
PRESENT-128
PRESENT-80
GIFT-64/128
SIMON-64/128
SPECK-64/128

Cpu


CPU Usage Comparison
CPU Utilization (%)
0
20
40
60
80
100
97.2%
97.3%
97.8%
98.1%
98.4%
99.7%
SIMON-64/128
SPECK-64/128
TinyJambu-128
PRESENT-128
GIFT-64/128
PRESENT-80

Latency


Latency vs Response Time by Algorithm
latency_time_ms
response_time_ms
Time (ms)
0
5
10
15
20
25
30
GIFT-64/128
PRESENT-128
PRESENT-80
SIMON-64/128
SPECK-64/128
TinyJambu-128
cipher

Attack Analysis Results

Attack	Target	Expected	Observed	Blocked	Status
Replay Attack	Packet Replay	Replay should be rejected	Replay attack detected	True	□ Protected
MITM Tampering	Data Integrity	Modified packet should fail MAC	MITM tampering result not available	True	□ Protected
Brute Force (Toy)	Toy 8-bit Key	Key should be recoverable	Recovered: True	False	□□ Demo Only
Brute Force (Real)	128-bit Keys	Should be infeasible	Theoretically infeasible	True	□ Resistant